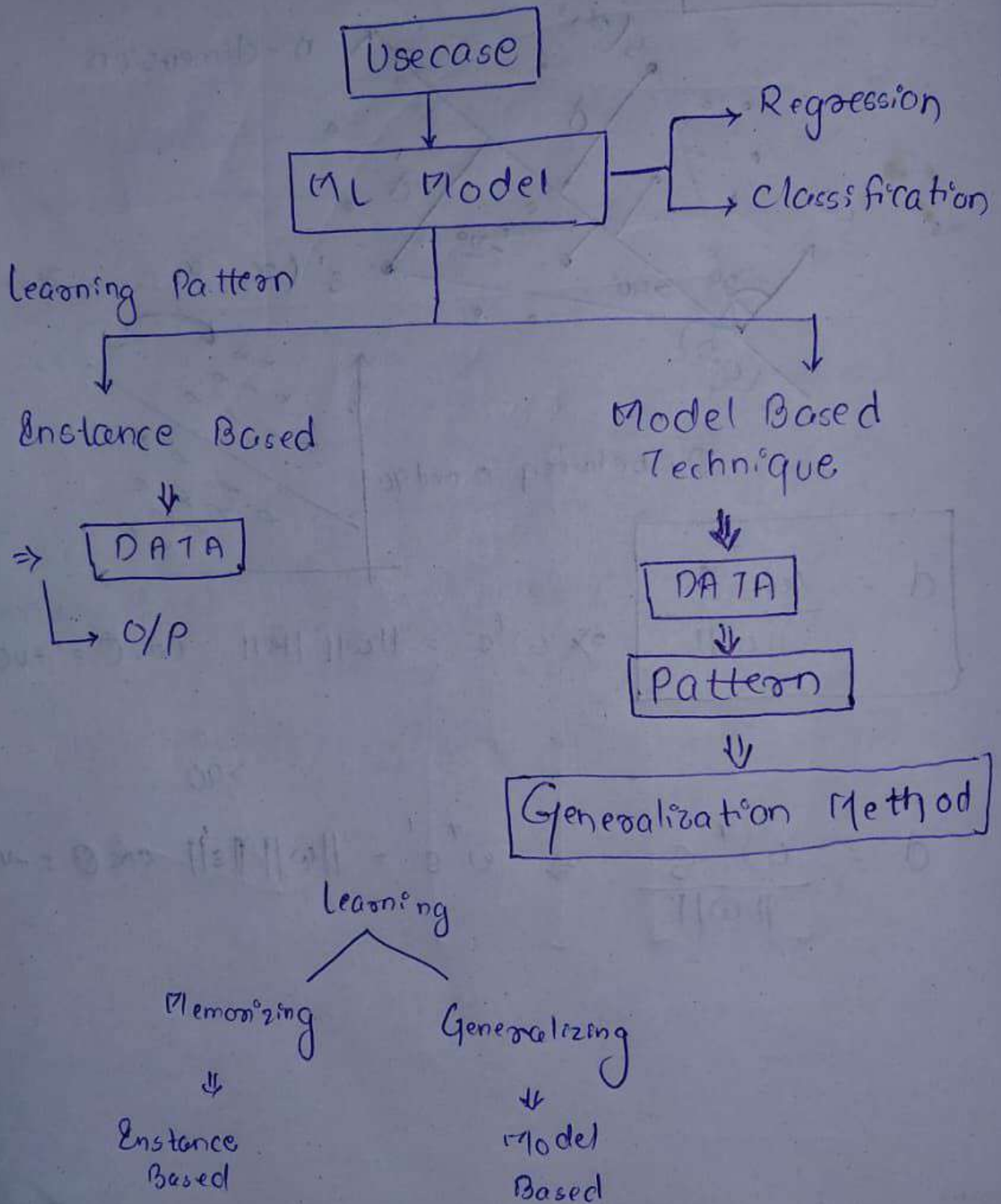


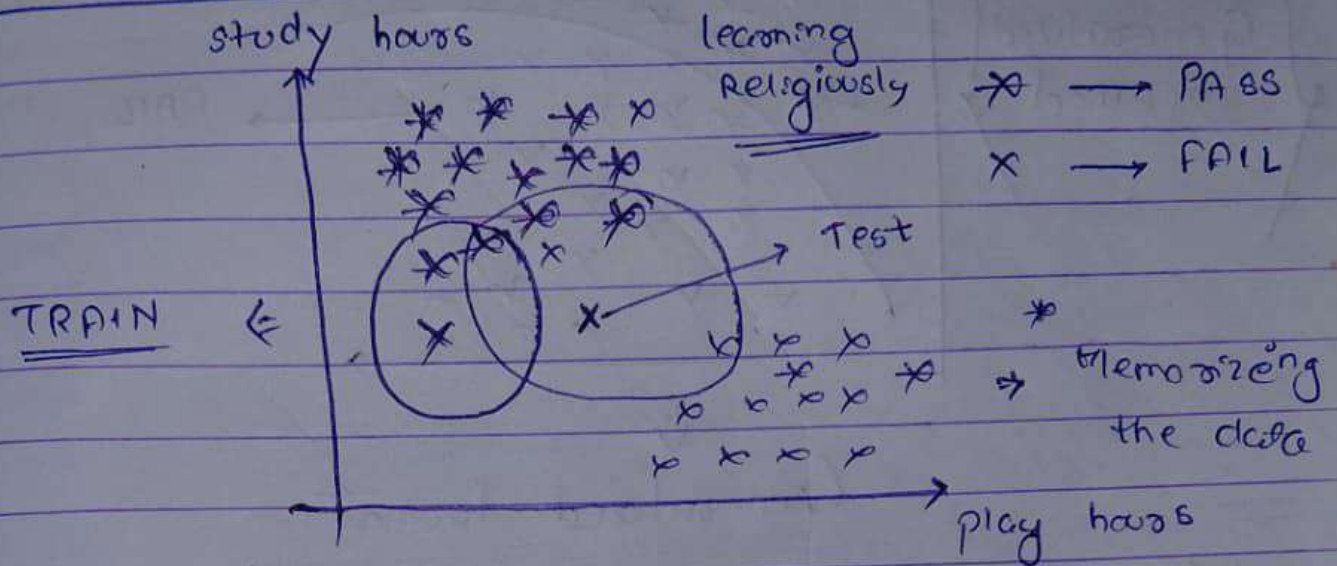
Instance Based Learning vs Model Based Learning



Instance Based

59

No. of Play hours	No. of study hours	Pass / Not
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Instance Based Learning



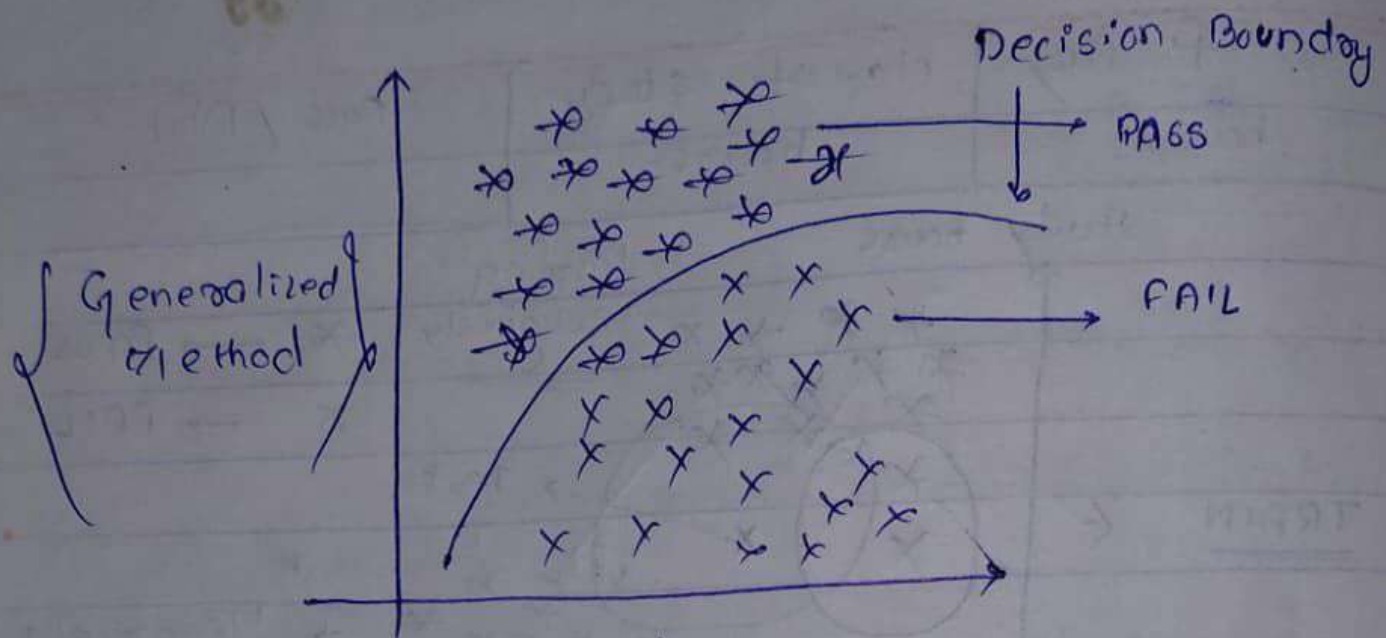
Domain Expert

(PASS/FAIL)

Notes: within a circle which has heigher point then this is the test output.

1) KNN (K Nearest Neighbour)

Model Based Technique



↓
Generalized format -

Note: Above the decision boundary the point is pass and below the decision boundary the point is fail.

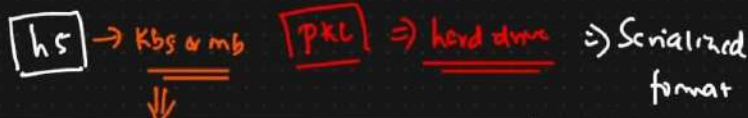
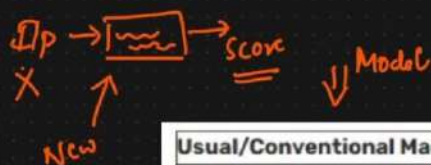
① learn pattern of the data

Usual/Conventional Machine Learning

- Prepare the data for model training.
- Train model from training data to estimate model parameters i.e. discover patterns.
- Store the model in suitable form.
- Generalize the rules in form of model, even before scoring instance is seen.
- Predict for unseen scoring instance using model.
- Can throw away input/training data after model training.
- Requires a known model form.
- Storing models generally requires less storage.
- Scoring for new instance is generally fast.

Instance Based Learning 59

- Prepare the data for model training. No difference here.
- Do not train model. Pattern Discovery postponed until scoring query received.
- There is no model to store.
- No generalization before scoring. Only generalize for each scoring instance individually as and when seen.
- Predict for unseen scoring instance using training data directly.
- Input/training data must be kept since each query uses part of full set of training observations.
- May not have explicit model form.
- Storing training data generally requires more storage.
- Storing for new instance may be slow.



Generalization
 Decision
 Boundaries

Usual/Conventional Machine Learning	Instance Based Learning
Prepare the data for model training ✓	Prepare the data for model training. No difference here ✓
Train model from training data to estimate model parameters i.e. <u>discover patterns</u>	Do not train model. Pattern discovery postponed until scoring query received
Store the model in suitable form	There is no model to store
Generalize the rules in form of model, even before scoring instance is seen	No generalization before scoring. Only generalize for each scoring instance individually as and when seen
Predict for unseen scoring instance using model	Predict for unseen scoring instance using training data directly
Can throw away input/training data after model training	Input/training data must be kept since each query uses part or full set of training observations
Requires a known model form	May not have explicit model form
Storing models generally requires less storage	Storing training data generally requires more storage
Scoring for new instance is generally fast	Storing for new instance may be slow

